

Note for Sieve Theory

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November 2023

1 LINEAR SIEVE

Theorem 1. *It is necessary for us to define the upper/lower bound sieve $\lambda^\pm$:*

$$\lambda^\pm(1) = 1, \quad (\lambda^- * u)(d) \leq 0 \leq (\lambda^+ * u)(d)$$

for all square-free d . In addition, $\lambda^\pm$ has support level D and sieving range $\mathcal{P}_0$ if and only if $\lambda^\pm(d) = 0$ for all $d \geq D$ and for all $p|d, p \notin \mathcal{P}_0$. A combinatorial sieve is said to have dimension $n > 0$ if

$$\prod_{\substack{p \in \mathcal{P} \\ y \leq p < z}} \frac{1}{1 - f(p)} \leq C \left(\frac{\log z}{\log y} \right)^n$$

for a constant C and $\forall 1 \leq y < z$. In case $n = 1$, it refers to linear sieve.

We first construct prime set $\mathcal{P}_0$ such that $\mathcal{Q} = \mathcal{P} \setminus \mathcal{P}_0$ is finite, Put

$$P_0(z) = \prod_{\substack{p \in \mathcal{P}_0 \\ p < z}} p, \quad Q(z) = \prod_{\substack{p \in \mathcal{Q} \\ p < z}} p, \quad Q = \prod_{p \in \mathcal{Q}} p; \quad |\mathcal{A}_d| = f(d)|\mathcal{A}| + r_d$$

$$F(z, \lambda^\pm) := \sum_{d|P_0(z)} \lambda^\pm(d) f(d)$$

then

$$|\mathcal{A}| F(z, \lambda^-) \prod_{d|Q(z)} (1 - f(p)) - \sum_{\substack{d < DQ \\ d|P(z)}} |r_d| \leq S(\mathcal{A}, \mathcal{P}, z) \leq |\mathcal{A}| F(z, \lambda^+) \prod_{d|Q(z)} (1 - f(p)) + \sum_{\substack{d < DQ \\ d|P(z)}} |r_d| \quad (1)$$

Lemma 1. *Define the sets*

$$\mathcal{D}^\pm := \left\{ \prod_{i=1}^j p_i \mid p_{i+1} < p_i < D; \forall i \equiv \frac{1}{2} \pm \frac{1}{2} \pmod{2}, \prod_{k=1}^{i-1} p_k \cdot p_i^3 < D \right\}$$

and function $\lambda_{D, \mathcal{P}}^\pm$

$$\lambda_{D, \mathcal{P}}^\pm(d) := \begin{cases} \mu(d) & d \in \mathcal{D}^\pm, d|P(D) \\ 0 & \text{otherwise} \end{cases}$$

Then we obtain that $\lambda_{D, \mathcal{P}}^\pm(d)$ are the upper and lower bound of sieves with support level D and sieving range $\mathcal{P}$, and $\forall d, |\lambda_{D, \mathcal{P}}^\pm(d)| \leq 1$.

Proof. Because $1 \in \mathcal{D}^\pm$, $\lambda_{D, \mathcal{P}}^\pm(1) = \mu(1) = 1$. By induction we can assume that $n|P(D)$.

If $\omega(n) = 1$ then n is a prime number such that $n \in \mathcal{P}, n < D$. Thus

$$\sum_{d|n} \lambda_{D, \mathcal{P}}^-(d) = \mu(1) + \mu(p) = 0$$

$$\sum_{d|n} \lambda_{D,\mathcal{P}}^+(d) = \mu(1) + \lambda_{D,\mathcal{P}}^+(p) \geq 0$$

If $\omega(n) = r + 1$, we write $n = \prod_{i=0}^r q_i$ with $q_i \in \mathcal{P}$. Set $N = \frac{n}{q_0}$. Thus by induction hypothesis

$$\sum_{d|N} \lambda_{D,\mathcal{P}}^-(d) \leq 0 \leq \sum_{d|N} \lambda_{D,\mathcal{P}}^+(d)$$

n 's divisors are either d or $q_0 d$ for $d|N$, so that

$$\begin{aligned} \sum_{d|n} \lambda_{D,\mathcal{P}}^+(d) &= \sum_{d|N} \lambda_{D,\mathcal{P}}^+(d) + \sum_{d|N} \lambda_{D,\mathcal{P}}^+(q_0 d) \geq \sum_{d|N} \lambda_{D,\mathcal{P}}^+(q_0 d) = \sum_{\substack{d|N \\ q_0 d \in \mathcal{D}^+}} \mu(q_0 d) = - \sum_{\substack{d|N \\ q_0 d \in \mathcal{D}^+}} \mu(d) \\ \sum_{d|n} \lambda_{D,\mathcal{P}}^-(d) &= \sum_{d|N} \lambda_{D,\mathcal{P}}^-(d) + \sum_{d|N} \lambda_{D,\mathcal{P}}^-(q_0 d) \leq \sum_{d|N} \lambda_{D,\mathcal{P}}^-(q_0 d) = \sum_{\substack{d|N \\ q_0 d \in \mathcal{D}^-}} \mu(q_0 d) = - \sum_{\substack{d|N \\ q_0 d \in \mathcal{D}^-}} \mu(d) \end{aligned} \quad (2)$$

We write $d = \prod_{i=1}^s p_i$ with $p_i \leq q_i < q_0 < D$. Consider $E = \frac{D}{q_0}$. It is better for us to construct a set $\mathcal{E}^\pm$ such that replace D with E in $\mathcal{D}^\pm$. Then $q_0 d \in \mathcal{D}^+$ if and only if $q_0^3 < D$ and $\prod_{j=1}^{i-1} p_j \cdot p_i^3 < E$ for every odd i . Hence we have

$$\sum_{\substack{d|N, q_0^3 \geq D \\ q_0 d \in \mathcal{D}^\pm}} \mu(d) = \sum_{\substack{d|N, q_0^3 \geq D \\ d \in \mathcal{E}^\pm}} \mu(d) = 0 \quad (3)$$

Since q_0 isn't qualified. Therefore

$$\sum_{\substack{d|N \\ q_0 d \in \mathcal{D}^+}} \mu(d) = \sum_{\substack{d|N, q_0^3 < D \\ q_0 d \in \mathcal{D}^+}} \mu(d) = \sum_{\substack{d|N, q_0^3 \geq D \\ d \in \mathcal{E}^+}} \mu(d) = \sum_{\substack{d|N \\ d \in \mathcal{E}^+}} \mu(d) \leq 0 \implies \sum_{d|n} \lambda_{D,\mathcal{P}}^+(d) \geq 0$$

Similarly, We can prove

$$\sum_{d|n} \lambda_{D,\mathcal{P}}^-(d) \leq 0$$

□

Lemma 2. For upper and lower bound function $\lambda_{D,\mathcal{P}}^\pm(d)$ with supporting level D and sieving range $\mathcal{P}$. Define

$$T_n(D, z) = \sum_{\substack{p_i \in \mathcal{P}, p_m < w_m \\ \forall 2 \nmid m < n \\ w_n \leq p_{i+1} < p_i < z}} f\left(\prod_{i=1}^n p_i\right) V(p_n) \quad (1 \leq i \leq n)$$

where w_n is defined with $w_n^2 \prod_{i=1}^n p_i = D$. It can be inferred

$$F(z, \lambda_{D,\mathcal{P}}^+(d)) = V(z) + \sum_{2 \nmid d} T_d(D, z) \quad F(z, \lambda_{D,\mathcal{P}}^-(d)) = V(z) - \sum_{2 \nmid d} T_d(D, z)$$

Proof. Consider $d|P(z)$ such that $d \in \mathcal{D}^+$ in which d is a product of distinct prime in $\mathcal{P}$ smaller than $z \leq D$ and for all odd i , $p_1 p_2 \dots p_i^3 < D$. Which we replace it with w_n , $p_i^2 \frac{D}{w_i^2} < D$. Hence $p_i < w_i$. Therefore we can rewrite

$$F(z, \lambda_{D,\mathcal{P}}^+(d)) = \sum_{\substack{p_i \in \mathcal{P}, p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^j f\left(\prod_{i=1}^j p_i\right) \quad (1 \leq i \leq j \leq n) \quad (4)$$

And for $V(z)$

$$\begin{aligned}
V(z) &= \sum_{d|P(z)} \mu(d) f(d) = \sum_{\substack{p_i \in \mathcal{P}, i \leq j \leq n \\ p_{i+1} < p_i < z}} (-1)^j f\left(\prod_{i=1}^j p_i\right) \quad (1 \leq i \leq j \leq n) \\
&= \sum_{\substack{p_i \in \mathcal{P}, p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^j f\left(\prod_{i=1}^j p_i\right) + \sum_{\substack{p_i \in \mathcal{P}, p_{i+1} < p_i < z \\ \exists 2 \nmid m < n, p_m \geq w_m}} (-1)^j f\left(\prod_{i=1}^j p_i\right) \\
&= F(z, \lambda_{D, \mathcal{P}}^+(d)) + \overbrace{\sum_{2 \nmid n} \sum_{\substack{p_i \in \mathcal{P}, p_n \geq w_n \\ p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^j f\left(\prod_{i=1}^j p_i\right)}^S
\end{aligned} \tag{5}$$

The inner sum S in (5) can be split as follows:

$$\begin{aligned}
\sum_{\substack{p_i \in \mathcal{P}, p_n \geq w_n \\ p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^j f\left(\prod_{i=1}^j p_i\right) &= \sum_{\substack{p_i \in \mathcal{P}, p_n \geq w_n \\ p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^n f\left(\prod_{i=1}^j p_i\right) \sum_{\substack{j \geq n+1, p_j \in \mathcal{P} \\ p_j < p_{i+1} < p_i < p_n}} (-1)^{j-n} f\left(\prod_{i=n+1}^j p_i\right) \\
&= \sum_{\substack{p_i \in \mathcal{P}, p_n \geq w_n \\ p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^n f\left(\prod_{i=1}^j p_i\right) \sum_{\substack{j \geq n+1, p_j \in \mathcal{P} \\ p_j < p_{i+1} < p_i < p_n}} \mu\left(\prod_{i=n+1}^j p_i\right) f\left(\prod_{i=n+1}^j p_i\right) \\
&= \sum_{\substack{p_i \in \mathcal{P}, p_n \geq w_n \\ p_{i+1} < p_i < z \\ \forall 2 \nmid m < n, p_m < w_m}} (-1)^n f\left(\prod_{i=1}^j p_i\right) V(p_n) = (-1)^n T_n(D, z)
\end{aligned} \tag{6}$$

Hence we obtained

$$V(z) = F(z, \lambda_{D, \mathcal{P}}^+(d)) - \sum_{2 \nmid d} T_d(D, z)$$

By analogous argument

$$V(z) = F(z, \lambda_{D, \mathcal{P}}^-(d)) + \sum_{2 \nmid d} T_d(D, z)$$

□

Theorem 2 (Jurkat-Richert). *Define the ϕ/Φ functions*

$$\Phi(x) := e^\gamma \left(u(x) + \frac{v(x)}{x} \right) \quad \phi(x) := e^\gamma \left(u(x) - \frac{v(x)}{x} \right)$$

Let $\mathcal{Q} \in \mathbb{P}$ be a finite set. If

$$\prod_{\substack{p \in \mathcal{P} \setminus \mathcal{Q} \\ y \leq p < z}} \frac{1}{1 - f(p)} \leq (1 + \epsilon) \frac{\log z}{\log y} \quad \mathcal{Q} = \prod_{p \in \mathcal{Q}} p$$

stands for some $0 < \epsilon < \frac{1}{200}$ and $\forall 1 < y < z$. Then $\forall D > z$

$$\left(\phi\left(\frac{\log D}{\log z}\right) - \epsilon e^{14 - \frac{\log D}{\log z}} \right) |\mathcal{A}| V(z) - R(D\mathcal{Q}, z) < S(\mathcal{A}, \mathcal{P}, z) < \left(\Phi\left(\frac{\log D}{\log z}\right) + \epsilon e^{14 - \frac{\log D}{\log z}} \right) |\mathcal{A}| V(z) + R(D\mathcal{Q}, z)$$

where the $R(D\mathcal{Q}, z)$ is

$$R(D\mathcal{Q}, z) = \sum_{\substack{d < D\mathcal{Q} \\ d|P(z)}} |r_d|$$